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SUPERSEDING
FRANKFORD ARSENAL
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6 May 1966

MILITARY SPECIFICATION

FIN ASSEMBLY; M149

1. SCOPE

1.1 This specification is the applicable detail specification for Fin Assembly, M149.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Cards
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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2.1 (Cont'd)

DRAWINGS
Munitions Command
D10520200 - Fin Assembly, M149
IEL-10520200 - Index of Inspection Equipment List for Fin Assembly, M149

PUBLICATIONS
ASA-B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Cartridge housing.- The cartridge housing shall have an unobstructed primer flash hole as shown on the applicable drawing.

3.3 Mechanical properties.- Mechanical properties shall be in accordance with the requirements of the applicable drawings and specifications.

3.4 Workmanship.- The requirements for workmanship are as shown on the applicable drawing, referenced specifications and the following:

3.4.1 Metal defective.- Soundness of the material shall be as required by the applicable material specifications.

3.4.2 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and shall be in accordance with the provisions of MIL-A-2550.

3.4.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.4.4 Foreign matter.- All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 15 metal parts assemblies as follows:

10 Fin assemblies, complete
5 Sets of component parts, unassembled

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Heat treatment lots.- Heat treatment lots shall contain cartridge housing forgings of the same alloy that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The heat treatment lot thus constituted shall be the basis for selecting the samples for the tension test hereinafter prescribed. The maximum number of cartridge housing forgings in a heat treatment lot shall consist of 15,000 or fraction thereof.

4.3.1.3 Bar stock lot.- Cartridge housings machined from bar stock shall consist of not more than 15,000 machined from the same alloy, temper and size of bar stock.

4.3.1.4 Extrusion lots.- A lot shall consist of fins from extruded stock made by the same process, of the same alloy and temper which has been aged in the same furnace charge.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for Major and Minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.3.2.a Control tool manufacture.- Dimensions marked "control tool manufacture" shall be gaged at the beginning of production whenever tooling is changed or as a minimum as defined below.

(a) 90 degree + 30 degree angle - Select and gage one housing from each machine at the start of each shift.

(b) Location of 20 flash holes - Select and gage one housing from each machine every two hours.

(c) Size of 20 flash holes - Select and gage one housing from each machine every two hours.

(d) Depth of primer flash hole (if applicable) - Select and gage one housing from each machine every two hours.

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4.3.2.a (Cont'd)

1. All material produced prior to periodic inspection for items (a), (b) and (c) is acceptable; however, corrective action must be taken whenever a defective characteristic is found.

2. All material produced subsequent to periodic inspection for item (d) will be rejected and screened whenever a defective characteristic is found and corrective action must be taken.

4.3.2.1 Holder, Increment - Drawing C7549026 covering a detail of Drawing D10520200

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Diameter, outside, major	Gage
202.	Diameter, outside, minor	Gage
203.	Diameter, inside	Gage
204.	Thickness	Gage
205.	Length of eye(s)	Visual (See 6.1)
206.	Width of eye(s)	Visual (See 6.1)
207.	Width of eye neck	Visual (See 6.1)
208.	Radii missing or incorrect	Visual (See 6.1)
209.	Surface finish improper	Visual (See 6.1)
210.	Eye(s) missing	Visual
211.	Metal defective	Visual
212.	Protective coating inadequate or damaged	Visual
213.	Burr	Visual
214.	Foreign matter	Visual

4.3.2.2 Housing Cartridge, Drawing C7549028 covering a detail of Drawing D10520200

CRITICAL

None defined

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity at front of taper with pitch diameter of body thread, including perpendicularity of front face	Gage
102. Concentricity at rear of taper with pitch diameter of body thread, including perpendicularity of front face	Gage
103. Cartridge cavity not functionally concentric with pitch diameter of body thread	Gage
104. Primer flash hole not functionally concentric with cartridge cavity	Gage
105. Pitch diameter of body thread, max.	Gage
106. Minor diameter of body thread, max.	Gage
107. Pitch diameter of fin thread, min.	Gage
108. Major diameter of fin thread, min.	Gage
109. Diameter of cartridge cavity	Gage
110. Diameter of primer flash hole	Gage
111. Diameter of primer flash hole counterbore (if applicable)	Gage
112. Depth to bottom of body thread cavity	Gage
113. Length of effective body thread, min.	Gage
114. Length of effective fin thread, min.	Gage
115. Flash hole(s) missing or excessive in number	Visual
116. Thread(s) damaged or not full	Visual
117. Metal defective	Visual
MINOR	
201. Pitch diameter of body thread, min.	Gage
202. Minor diameter of body thread, min.	Gage
203. Diameter at forward basic length	Gage
204. Diameter at rear basic length	Gage
205. Diameter of body thread counterbore	Gage
206. Depth of body thread counterbore	Gage
207. Depth to bottom of cartridge cavity	Gage
208. Length from rear face to fin seat	Gage
209. Length, total	Gage
210. Radii or chamfer missing or incorrect	Visual (See 6.1)
211. Surface finish improper	Visual (See 6.1)
212. Burr	Visual
213. Foreign matter	Visual

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4.3.2.3 Fin - Drawing D10520199 covering a detail of Drawing D10520200

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of fin blades with pitch diameter of primer thread, including perpendicularity of primer thread counterbore face	Gage
102. Primer thread counterbore not functionally concentric with pitch diameter of primer	Gage
103. Straightness of fin blades, including twist after canting thread	Gage
104. Pitch diameter of housing thread, max.	Gage
105. Minor diameter of housing thread, max.	Gage
106. Pitch diameter of primer thread, max.	Gage
107. Minor diameter of primer thread, max.	Gage
108. Pitch diameter of through thread, max. (if applicable)	Gage
109. Minor diameter of through thread, max. (if applicable)	Gage
110. Diameter of housing thread counterbore	Gage
111. Diameter of fin blades (large)	Gage
112. Diameter of fin blades (small)	Gage
113. Diameter of primer thread counterbore (large)	Gage
114. Diameter of primer thread counterbore (small)	Gage
115. Depth of housing thread counterbore	Gage
116. Depth of primer thread counterbore	Gage
117. Length of effective housing thread, min.	Gage
118. Length of effective primer thread, min.	Gage
119. Length from front face to bottom of large rear counterbore	Gage
120. Cant of fin blades incorrect	Gage
121. Tangency of radius with hub diameter and fin blade surface	Visual (See 6.1)
122. Fin blades canted in improper direction	Visual
123. Thread(s) damaged or not full	Visual
124. Metal defective	Visual

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Angle at front of fin blades	Gage
202.	Angle between fin blades	Gage
203.	Pitch diameter of primer thread, min.	Gage
204.	Minor diameter of primer thread, min.	Gage
205.	Pitch diameter of through thread, min. (if applicable)	Gage
206.	Minor diameter of through thread, min. (if applicable)	Gage
207.	Diameter of hub	Gage
208.	Diameter inside between threads (if applicable)	Gage
209.	Length of cant	Gage
210.	Length, total	Gage
211.	Thickness of fin blades	Gage
212.	Radii or chamfers missing or incorrect	Visual (See 6.1)
213.	Surface finish improper	Visual (See 6.1)
214.	Burr	Visual
215.	Foreign matter	Visual

4.3.2.4 Fin Assembly - Drawing D10520200

CRITICAL		
1.	Primer flash hole missing or obstructed	Visual
MAJOR		
101.	Outside diameter of fin blades not functionally concentric with pitch diameter of body thread, including perpendicularity with front face	Gage
102.	Minor diameter of primer thread not functionally concentric with pitch diameter of body thread including perpendicularity with front face (if applicable)	Gage
103.	Length, total, max.	Gage
104.	Fin blades distorted or damaged	Visual (See 6.1)
105.	Increment holder missing or damaged	Visual
106.	Thread(s) damaged or not full	Visual
107.	Metal defective	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Marking missing, incorrect or illegible	Visual
202. Burr	Visual
203. Foreign matter	Visual

4.3.2.5 Packed Carton prior to sealing

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Support(s) missing or damaged	Visual
202. Divider(s) missing or damaged	Visual
203. Improperly packed	Visual

4.3.2.6 Packed Carton after sealing

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual
202. Cartons damaged (cut or damaged through all layers)	Visual
203. Carton not properly sealed	Visual
204. Marking missing, incorrect or illegible	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with provisions of 4.4.

4.3.3.1 Mechanical properties.

4.3.3.1.1 Hardness.

4.3.3.1.1.1 Heat treatment lots.- In order to determine uniformity of heat treatment, 80 cartridge housing forgings in the heat treatment lot (as defined by 4.3.1.2) shall be subjected to a hardness test with a Brinell hardness tester or by any other method approved by the procuring activity.

4.3.3.1.1.2 Bar stock lot.- Eighty cartridge housings from the bar stock (as defined by 4.3.1.3) shall be subjected to a hardness test, with a Brinell hardness tester or by any other method approved by the procuring activity.

4.3.3.1.2 Tension test.

4.3.3.1.2.1 Heat treatment lots.- Two cartridge housings, one of the hardest and one of the softest as determined by 4.3.3.1.1.1 shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.1.2.2 Bar stock lots.- Two cartridge housings, one of the hardest and one of the softest (as determined by 4.3.3.1.1.2) shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot.

4.3.3.1.2.3 Extrusion lots.- From the extrusion lot (as defined by 4.3.1.4) one fin from the first 1000 feet or fraction thereof shall be subjected to this test. One additional fin from each 2000 feet or fraction thereof for the remainder of the extrusion lot, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirement of the applicable drawing shall be cause for rejection of the lot.

4.3.3.2 Torque test.- Five fin assemblies from each one hour's production or fraction thereof from each torque machine in operation shall be selected at random and tested to determine compliance with the applicable drawing. Failure of any fin assembly to comply shall be cause for rejection of the production represented by the sample.

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4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.

4.4.1.1.1 Heat treatment lots.-Prior to finish machining, the samples supplied in accordance with 4.3.3.1.1.1 will be subjected to a hardness test on either end of the cartridge housing. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151. A minimum dwell time of 5 seconds shall be used.

4.4.1.1.2 Bar stock lots.-The samples supplied in accordance with 4.3.3.1.1.2 will be subjected to a hardness test on either end of the cartridge housing. The test shall be performed using the equipment and procedures prescribed by Federal Test Method Standard No. 151. A minimum dwell time of 5 seconds shall be used.

4.4.1.2 Tension test.

4.4.1.2.1 Cartridge housings.-From the cartridge housings selected in accordance with 4.3.3.1.2.1 or 4.3.3.1.2.2, one tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally. The number of specimens is for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens if necessary to determine that the mechanical properties of the cartridge housing meet the drawing requirements. The criteria for performing the tests shall be as prescribed by Federal Test Method Standard No. 151.

4.4.1.2.2 Fins.-From the samples selected in accordance with 4.3.3.1.2.3, one tension test specimen conforming to the standard "F2" type specimen shall be taken longitudinally from a fin blade. The number of specimens is for convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens if necessary to determine that the mechanical properties of fin meet the drawing requirements. The criteria for performing the tests shall be as prescribed by Federal Test Method Standard No. 151.

4.4.2 Torque test.-The samples supplied in accordance with 4.3.3.2 shall be tested by applying the required assembly torque to the fin in the direction of assembly before the thread sealing compound has set. Any perceptible movement of any fin will be considered as failure of the assembly to comply with the drawing requirements.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Packaging, packing and marking shall be in accordance with the requirements of level C in MIL-STD-1169.

5.2 Data card.-Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

CUSTODIAN:

Army - MU

PREPARING ACTIVITY:

Army - MU

Project No. 1315 - A626

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